Contrast Prediction for Fractal Image Compression

Shou-Cheng Hsiung and J. H. Jeng
Department of Information Engineering
I-Shou University, Kaohsiung County, Taiwan
d9303004@stmail.isu.edu.tw, jjeng@isu.edu.tw

Abstract

In this paper, a fast encoding algorithm is developed for Fractal Image Compression. At each search entry in the domain pool, by using the predicted contrast coefficients, we simplify the calculations for the contrast and brightness parameters. The redundant computations of eight dihedral symmetries of the domain block are also eliminated in the new encoding algorithm. Experimental results show that the encoding time is about two times faster than that of the full search method with almost the same PSNR for the retrieved image.

1 Introduction

The idea of the Fractal image compression is based on the assumption that the image redundancies can be efficiently exploited by means of block self-affine transformations. Many authors [1-3] suggest that at a block level images contain a large amount of self-similarity, and that the Fractal transform coding can be used to take advantage of this fact. The Fractal transform for image-data compression was introduced first by M. F. Barnsley and S. Demko [4]. However, the algorithm requires extensive computations [5]. The first practical Fractal image compression scheme was introduced by A. E. Jacquin [6], E. W. Jacobs, R. D. Boss and Y. Fisher [7] called the Jacquin-Fisher algorithm using block-based transformations and an exhaustive search strategy. Their approach was an improved version of the system patented by Barnsley [8,9]. The computational complexity was reduced by partitioning and classifying the image sub-blocks.

There are two important issues which need to be improved in the Jacquin-Fisher algorithm. First, the cost of an exhaustive search of a pool of domain blocks is too high. Secondly, a good classification algorithm needs to be obtained. Such a classification algorithm must be able to reduce the search space and also to decrease the order of the Fractal function. The recent work toward these efforts is reported in the book [10]. The Quadtree Fractal-Encoding scheme, developed by Y. Fisher [10], is a well-known technique that uses the recursive partition of the image into non-overlapping ranges with tree structure and the quadrant classification of both domain and range blocks. These techniques relate to a more general method called the classified vector quantization [11]: compute the first and second moments of the domain blocks and the orientation of their sub-blocks to find the visual distinction between blocks, to thereby reduce the domain-search space.

For the methods mentioned above, there is a need to consider the eight orientations of the allocated domain blocks. These are known as the dihedral operations on the block. This group of eight operations involve extensive computations in order to realize the above encoding procedure. It is well known [12-14] that a fast transform method can be used to implement the DCT, called the fast DCT. By performing the MSE computations in the frequency domain, after the proper arrangement, it follows that all of the redundant computations can be eliminated.

Since most of the energy of a block is located in the low-band data, we develop a DCT-like transformation in spatial domain to acquire only three coefficients to represent the block. This reduces the complexity of the inner product computations. Also, the eight operations of the Dihedral group can be directly obtained by changing the signs of the three DCT-like coefficients. Therefore, the purpose of speedup can be achieved.

2 Fractal coding

Fractal image compression is an inverse problem, i.e., for the given set S_W, find the IFS which has S_W as its attractor. It should be noted that, when S_W is a natural image, such IFS can hardly exist because most of the sub-images are not directly similar to the whole image. To solve this problem, the idea of local self-similarity is adopted to form the Partitioned Iterated Function System (PIFS) in which the contractive maps w_i is defined only on D_i where $D_i \subset X$ for $i = 1,...,n$.

For practical implementation, let f be a given

256×256 gray level image. The domain pool D is defined as the set of all possible blocks of size 16×16 of the image f, which makes up $(256-16+1) \times (256-16+1) = 58081$ blocks. The range pool R is defined to be the set of all non-overlapping blocks of size 8×8, which makes up $(256/8) \times (256/8) = 1024$ blocks. For each block v from the range pool, the fractal transformation is constructed by searching in the domain pool D the most similar block. At each search entry, the domain block is sub-sampled such that it has the same size as the range block. Then, the sub-sampled block is transformed subject to the eight transformations in the Dihedral group on the pixel positions. Assuming the origin of the image block is located at the center, the eight transformations $T_k : k = 0, \ldots, 7$ can be represented by the following matrices:

$$T_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad T_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$T_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad T_3 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

$$T_4 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad T_5 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

$$T_6 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad T_7 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

The transformations T_1 and T_2 correspond to the flips along the vertical and horizontal lines, respectively. T_3 is the flip along both the vertical and horizontal lines. T_4, T_5, T_6 and T_7 are the transformations of T_0, T_1, T_2 and T_3, respectively, performed by an additional flip along the main diagonal line.

Let u denote a sub-sampled domain block of the same size as v. The similarity of u and v is measured using Mean Square Error (MSE) defined by

$$d(u, v) = \frac{1}{64} \sum_{j=0}^{7} \sum_{i=0}^{7} (u(i, j) - v(i, j))^2 \qquad (1)$$

In another word, for each range block R_i, the fractal encoder searches from the set of domain blocks a domain block D_i and a contractive mapping w_i which minimizes $d(R_i, w_i(D_i))$ where

$$w_i \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ q \end{bmatrix}. \qquad (2)$$

The transformations w_i are also called the affine transformation. The parameters (*a, b, c, d*) constitute the eight Dihedral transformations, (t_x, t_y) is the position of the domain block, p is the contrast scale factor and q is the luminance offset. In the minimization process, p and q can be computed directly as

$$p_k = \frac{\left[N < u_k, v > - < u_k, \vec{1} >< v, \vec{1} >\right]}{\left[N < u_k, u_k > - < u_k, \vec{1} >^2\right]} \qquad (3)$$

and

$$q_k = \frac{1}{N}\left[< v \cdot \vec{1} > - p < u_k \cdot \vec{1} >\right] \qquad (4)$$

where $N = 64$ and $\vec{1} = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T$.

After the appropriate affine transformations was found for all range blocks, the parameters t_x, t_y, p, q and the orientation of eight Dihedral transformations are stored, which are referred to as the fractal codes.

In the decoding phase, one chooses an arbitrary initial image, and then uses the fractal codes to compute the attractor of each transformation w_i. After appropriate times of iterations, the image can be reconstructed, which has some degree of loss corresponding to the original image.

3 The proposed algorithm

In equation (3), the range block v and domain block u_k consist of 64 pixel values in spatial domain. Inspired from the fast DCT algorithm, we develop a new simple transformation to spare the cost of DCT computation. We first derived the mean values of the four sub-blocks, as shown in figure 1. By manipulating the four mean values, we can get three DCT-like coefficients.

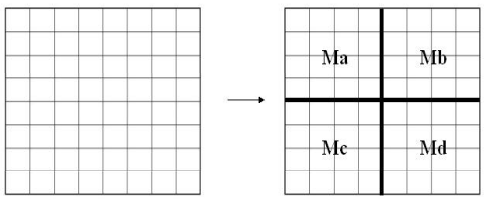


Figure 1. Mean values of the four sub-blocks.

Let b be a given 8×8 image block. The DCT-like transform of b is a 3-dim vector $\hat{b} = (\beta_1, \beta_2, \beta_3)$ given by

$$\begin{aligned} \beta_1 &= (M_a + M_c) - (M_b + M_d) \\ \beta_2 &= (M_a + M_b) - (M_c + M_d) \\ \beta_3 &= (M_a + M_d) - (M_b + M_c) \end{aligned} \qquad (5)$$

The DCT-like coefficients of $\hat{b}_k$, where $b_k, k = 0..7$ are the Dihedral transformation of b, can be easily computed as

$$b_0 = (\beta_1, \beta_2, \beta_3), \quad b_1 = (-\beta_1, \beta_2, \beta_3),$$

$$b_2 = (\beta_1, -\beta_2, \beta_3), \quad b_3 = (-\beta_1, -\beta_2, \beta_3),$$

$$b_4 = (\beta_2, \beta_1, \beta_3), \quad b_5 = (-\beta_2, \beta_1, \beta_3),$$

$$b_6 = (\beta_2, -\beta_1, \beta_3), \quad b_7 = (-\beta_2, -\beta_1, \beta_3)$$

As a consequence, there is no need to separately calculate the DCT-like coefficients of the 8 Dihedral transformations.

Since these coefficients are the predicted values for the DCT low-band data. Equations (3) for the contrast parameter has changed to

$$p_k = <\hat{v},\hat{v}>/<\hat{u}_k,\hat{v}> \qquad (6)$$

where $\hat{v}$ and $\hat{u}_k$, the transformed blocks, consist of only three integers. The steps to construct the transformed blocks are given as follows.

Step1. Divide each 8×8 block into four blocks.

Step2. Compute the four mean values.

Step3. Derive the three frequency-like parameters.

Step4. Form the transformed block.

As shown previously, the Dihedral transformed domain blocks needs only the three coefficients. This reduces the complexity of inner product in equation (6). Consequently, our fast encoding algorithm requires less computations than the original algorithm.

4 Experimental Results

To emphasize the encoding speed with a constant quality, the fast algorithm is compared to the baseline algorithm having the same compression ratio. The data is encoded using 8 bits for the translations t_x and t_y, respectively, 7 bits for the brightness q, 5 bits for the contrast p and 3 bits for the dihedral transformation. A total of 31 bits are needed for coding each range block. The PSNR given by $PSNR = 10\cdot\log_{10}(255^2 / MSE)$ is used to measure the quality of the retrieved images. For the different algorithms, the retrieved image is obtained by the same number of iterations of the encoded affine transformation with the same initial image.

The software simulation is done using C++ Builder 5 on a Pentium 2.4G, windows XP pc. The tested pattern is the gray level Lena and Peppers images of size 256×256. Figures 2-4 shows the decoded images by using the full search method and the proposed algorithm. Table 1 gives the encoding time for finding p and q. The proposed algorithm is about two times faster than that of the full search method with almost the same PSNR.

Table 1. The comparison of full search and proposed methods.

	Method	PSNR	Time(sec) For p+q
Lena	Full Search	29.18	325
	Proposed	28.99	171
Peppers	Full Search	29.86	326
	Proposed	29.72	175
Baboon	Full Search	21.81	325
	Proposed	21.73	172

(a)

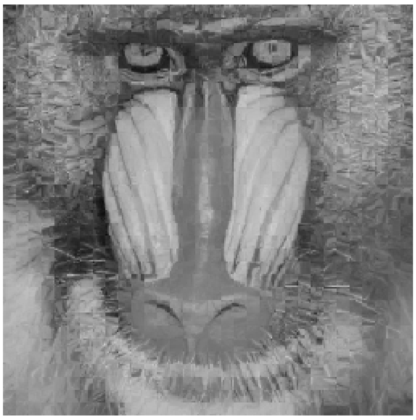
(b)

Figure 2. (a) Full search method decoded Lenna, PSNR=29.18
(b) Proposed method decoded Lenna , PSNR=28.99

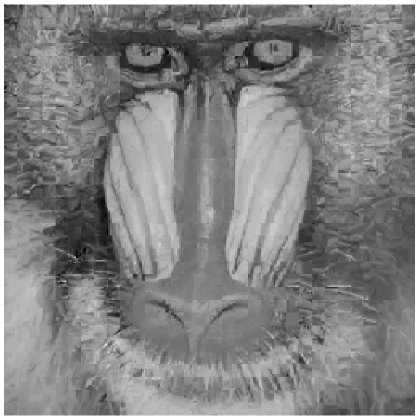
(a)

(b)

Figure 3. (a) Full search method decoded Peppers, PSNR=29.86
(b) Proposed method decoded Peppers, PSNR=29.72



(a)



(b)

Figure 4. (a) Full search method decoded Baboon, PSNR=21.81
(b) Proposed method decoded Baboon, PSNR=21.73

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